

D2_Calcium_Waves_1D

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1 D2 · Calcium Waves in 1-D — Travelling Pulses in an Excitable Medium

Module D — Spatio-Temporal Patterns

In D1 we met the calcium–IP3 model and explored its single-cell dynamics: steady states, oscillations, and excitability near the Hopf bifurcation.

Now we line up 80 cells in a chain, couple them by calcium diffusion, and watch what happens when a small pacemaker region continuously fires.

The result is a **travelling wave** — a periodic calcium pulse that propagates from the pacemaker through the rest of the tissue. This is the basic mechanism behind calcium waves in *Xenopus* oocytes, fertilisation waves in eggs, and intercellular calcium signalling in astrocyte networks.

Learning goals - Extend the single-cell IP3 ODE to a spatially extended chain. - Understand the difference between an **oscillatory** medium (all cells fire spontaneously) and an **excitable** medium (cells fire only when triggered). - Read a kymograph and extract wave speed and period.

1.1 1 · From Single Cell to Chain

1.1.1 Single-cell model (D1 review)

$$\dot{X}_i = a_i - \frac{m_2 X_i}{1 + X_i} + \frac{m_3 Y_i}{k_1 + Y_i} \cdot \frac{X_i^2}{k_a + X_i^2} + Y_i - k X_i$$

$$\dot{Y}_i = \frac{m_2 X_i}{1 + X_i} - \frac{m_3 Y_i}{k_1 + Y_i} \cdot \frac{X_i^2}{k_a + X_i^2} - Y_i$$

X = cytosolic Ca^2 , Y = IP .

1.1.2 Coupling — calcium diffusion

Calcium spreads between neighbouring cells (through gap junctions or ER continuity):

$$\dot{X}_i = f(X_i, Y_i) + D_X \underbrace{(X_{i-1} - 2X_i + X_{i+1})}_{\text{discrete Laplacian}}$$

Only calcium (X) diffuses; IP (Y) is produced and consumed locally.

1.1.3 Oscillatory vs excitable medium

From D1's bifurcation scan:

Inflow a	Single-cell behaviour
$a < 0.32$	Stable steady state — excitable (fires only when triggered)
$a \geq 0.32$	Limit cycle — oscillatory (fires spontaneously)

We set the **bulk** cells just below threshold ($a = 0.31$, excitable) and the **pacemaker** cells above it ($a = 0.34$, oscillating).

The pacemaker fires periodically and each calcium pulse triggers its excitable neighbours — which trigger their neighbours — propagating a wave. Unlike phase waves in a purely oscillatory medium, these are **true propagating fronts**: each cell fires, enters a refractory period, and cannot be retriggered until it recovers.

```
[1]: import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
from scipy.signal import find_peaks

# Model parameters (identical to D1)
PARAMS = dict(m2=20, m3=23, ka=0.8, k=0.8, k1=0.8)

def calcium_ip3(t, state, a, m2, m3, ka, k, k1):
    X, Y = state
    ip3 = (m3 * Y / (k1 + Y)) * X**2 / (ka + X**2)
    dX = a - m2*X/(1+X) + ip3 + Y - k*X
    dY = m2*X/(1+X) - ip3 - Y
    return [dX, dY]

print("Model loaded. Oscillatory threshold: a 0.32")
```

Model loaded. Oscillatory threshold: a 0.32

1.2 2 · Chain Setup

```
[2]: # Chain parameters
N = 80 # number of cells
D_X = 1.0 # calcium diffusion coefficient
N_PACE = 5 # pacemaker cells at the left end
A_PACE = 0.35 # pacemaker inflow (oscillatory, period 10)
A_BULK = 0.31 # bulk inflow (excitable, below threshold)

# ODE for the full chain
def chain_rhs(t, y, N, a_arr, p, D_X):
    X, Y = y[:N], y[N:]
```

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ip3    = (p['m3']*Y/(p['k1']+Y)) * X**2/(p['ka']+X**2)
dX     = a_arr - p['m2']*X/(1+X) + ip3 + Y - p['k']*X
dY     =      p['m2']*X/(1+X) - ip3      - Y
Xpad   = np.pad(X, 1, mode='edge')          # no-flux boundaries
dX     += D_X * (Xpad[:-2] - 2*X + Xpad[2:])
return np.concatenate([dX, dY])

# Spatially varying inflow
a_arr = np.full(N, A_BULK)
a_arr[:N_PACE] = A_PACE

# Initial conditions: all cells at the excitable steady state
y0 = np.concatenate([np.full(N, 0.48), np.full(N, 2.54)])

print(f"Chain: {N} cells")
print(f"Pacemaker: cells 0-{N_PACE-1} (a = {A_PACE}, oscillatory)")
print(f"Bulk:      cells {N_PACE}-{N-1} (a = {A_BULK}, excitable)")
print(f"Calcium diffusion: D_X = {D_X}")

```

```

Chain: 80 cells
Pacemaker: cells 0-4 (a = 0.35, oscillatory)
Bulk:      cells 5-79 (a = 0.31, excitable)
Calcium diffusion: D_X = 1.0

```

```

[3]: T_END = 200
print(f"Integrating {N} cells for t = 0 ... {T_END} ...")

sol = solve_ivp(
    chain_rhs, (0, T_END), y0,
    args=(N, a_arr, PARAMS, D_X),
    method='LSODA', dense_output=True,
    rtol=1e-6, atol=1e-8,
)
print(f"Done. {sol.message}")

# Evaluate on a regular grid
N_T = 2000
t_eval = np.linspace(0, T_END, N_T)
Y_out = sol.sol(t_eval)
X_xt = Y_out[:,N, :]      # calcium: shape (cells, timepoints)

```

```

Integrating 80 cells for t = 0 ... 200 ...
Done. The solver successfully reached the end of the integration interval.

```

1.3 3 · Kymograph

Each row is one cell; colour is Ca^{2+} concentration. Bright diagonals = wave fronts travelling from left (pacemaker) to right (bulk).

- **Stripe slope** → wave speed (cells per time unit).
- **Stripe spacing along time axis** → oscillation period 10.
- The first wave takes time to cross the chain; subsequent waves follow steadily.

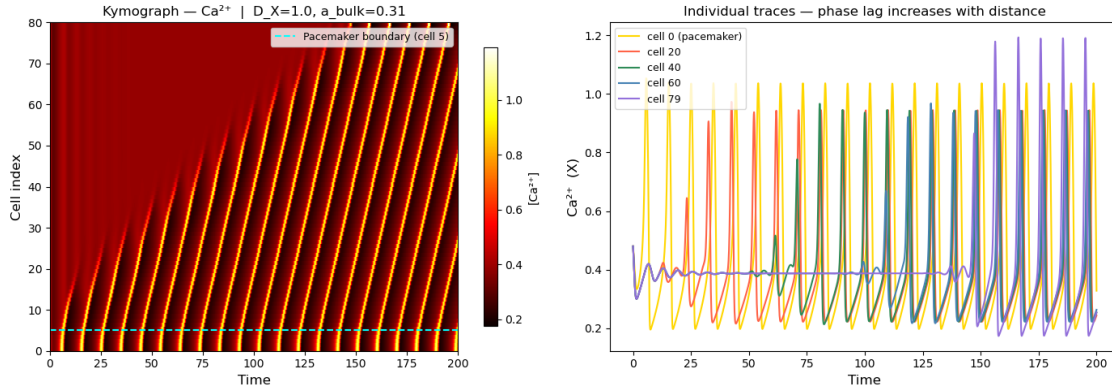
```
[4]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: kymograph
ax = axes[0]
im = ax.imshow(X_xt, aspect='auto', origin='lower',
               extent=[0, T_END, 0, N],
               cmap='hot', interpolation='nearest',
               vmin=X_xt.min(), vmax=X_xt.max())
ax.axhline(N_PACE, color='cyan', lw=1.5, ls='--',
          label=f'Pacemaker boundary (cell {N_PACE})')
ax.set_xlabel('Time', fontsize=12)
ax.set_ylabel('Cell index', fontsize=12)
ax.set_title(f'Kymograph - Ca2 | D_X={D_X}, a_bulk={A_BULK}', fontsize=12)
ax.legend(fontsize=9)
plt.colorbar(im, ax=ax, label='[Ca2]', shrink=0.85)

# Right: individual traces
ax2 = axes[1]
for ci, col in zip([0, 20, 40, 60, 79],
                  ['gold', 'tomato', 'seagreen', 'steelblue', 'mediumpurple']):
    label = f'cell {ci}' + (' (pacemaker)' if ci < N_PACE else '')
    ax2.plot(t_eval, X_xt[ci], color=col, lw=1.5, label=label)
ax2.set_xlabel('Time', fontsize=12)
ax2.set_ylabel('Ca2 (X)', fontsize=12)
ax2.set_title('Individual traces - phase lag increases with distance',
             ↵fontsize=12)
ax2.legend(fontsize=9)

plt.tight_layout()
plt.show()

# Wave speed estimate
# Time for first peak to travel from cell 0 to cell 79
pk0, _ = find_peaks(X_xt[0], prominence=0.2)
pk79, _ = find_peaks(X_xt[79], prominence=0.2)
if len(pk0) > 0 and len(pk79) > 0:
    travel_time = t_eval[pk79[0]] - t_eval[pk0[0]]
    speed = (79 - 0) / travel_time
    period = np.mean(np.diff(t_eval[pk0])) if len(pk0) > 1 else float('nan')
    print(f"First wave: cell 0 → cell 79 in {travel_time:.1f} time units")
    print(f"Wave speed: {speed:.2f} cells / time unit")
    print(f"Pacemaker period: {period:.1f} time units")
```



First wave: cell 0 → cell 79 in 141.5 time units

Wave speed: 0.56 cells / time unit

Pacemaker period: 9.6 time units

1.4 4 · Exploration

Return to the **Chain parameters** cell and try the following.

Experiment	Change	What to observe
Faster diffusion	D_X = 5.0	Wave travels faster; stripes steepen
Slower diffusion	D_X = 0.5	Wave travels slower or may fail to propagate
All oscillatory	A_BULK = 0.33	All cells fire spontaneously → synchronisation
Further below threshold	A_BULK = 0.28	Bulk less excitable; wave may die out
Larger pacemaker	N_PACE = 15	Wider initiation zone; same wave speed
Two pacemakers	a_arr[75:] = 0.34 after setting up	Waves collide and annihilate

Two-pacemaker collision is particularly informative: unlike sound waves, calcium wave fronts annihilate on collision because both media enter refractory state simultaneously. This is a hallmark of excitable-medium waves.

1.5 Summary

Feature	Calcium wave (excitable)	Phase wave (oscillatory)
Requires pacemaker?	Yes	Yes (without symmetry breaking: synchrony)
Wave speed	Set by diffusion + kinetics	Set by phase gradient

Feature	Calcium wave (excitable)	Phase wave (oscillatory)
Collision	Annihilation	Pass through each other
Refractory period	Yes — cells cannot re-fire immediately	No

The excitable mechanism produces **robust, unidirectional** wave trains — exactly the behaviour observed in biological calcium signalling.

Connection to D3: placing the same model on a 2-D grid with a circular pacemaker produces **target patterns** (concentric rings). Breaking the rotational symmetry of the pacemaker generates **spiral waves** that rotate indefinitely — the spectacular patterns seen in *Dictyostelium* and cardiac tissue.